Exploring Melbourne House Prices (2016–2017)

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Originally I used the Melbourne housing dataset, which contained thirty five thousand unique values, but after I finished the project, I realized I had used the wrong dataset, and when I finally resolved the issue, the dataset I used was melbourne housing "snapshot" (notice how its a one word difference and the dataset is originally one and the same?) containing almost fourteen thousand unique values, thousands of residential property sales across Melbourne, Australia. Each column includes features describing the properties size, location, type, and neighbourhood. Some of the questions I had were, What features of a property, like size or location, make it sell for a higher price?, Do properties closer to the city center always cost more, or are there cheaper suburbs that surprise us? Which property type gives you the most space and rooms for the price you pay? How does the number of bedrooms and bathrooms affect the price of a property? Among others. My main goal was to understand what makes Melbourne properties more expensive than others. To do this i looked at features like size, number of rooms and location affect the final sale price, whether properties closer to the city center tend to cost more than those further out, how much space and how many rooms you get depending on the type of property you buy,and whether having more bedrooms and bathrooms leads to a noticeable higher price tag.

1. Cleaning

The raw data I had was riddled with problems that needed fixing before I could move onto analysis so I needed to clean it to prepare. Three columns had too many missing values to be useful, buildingarea, yearbuilt, and date, so I entirely got rid of them, as imputing that many values wouldnt have been ideal. The other columns however (distance, postcode, councilarea, propertycount) I only dropped the rows where values were missing, using mean or median imputation depending on the shape of the distribution. The columns that were right skewed(Landsize, bathroom) is used the median, which is more robust to outliers, and columns like latitude and longitude were symmetric so i used mean. Several columns were stored as the wrong type when first loading the dataset, date was stored as plain text, so i converted it to a proper datetime. Buildingarea, yearbuilt and propertycount were also stored as text in rare cases, so I decided to convert them to numeric values, also using errors="coerce", making sure I turn anything unreadable into Nan instead of plainly crashing. For the outliers i used the iqr method on the price column to detect extreme values, and anything priced above the ninety-ninth percentile was capped using clip.(), as to not distort the analysis. For the duplicates i simply removed any duplicated columns as to make sure no property sale was counted more than once. I then wrapped all of these steps into a clean_data() function and added three Final checks to make sure there were no missing values in price, all prices were greater than zero and a final column number of eighteen.

2. Feature engineering

I needed to capture information raw columns could not, so i created various new features.

- bedroom_x_bathroom = Bedroom2 × Bathroom

This interaction feature explains the idea that a property with more bedrooms and bathrooms is worth more than either alone. For example, a property with 4 bedrooms and 3 bathroom sis likely more valuable than one with 4 bedrooms and only one bathroom. Multiplying the 2 into a single number would help the model better pick up on that relationship

- Encoding categorical columns

The Type column(describes whether a property is a house, a unit or a townhouse) and the method column(explaining how each property was sold) were both converted into numbers using one hot encoding turning each category into its own column of ones and zeros, so the model can actually understand them, using dropfirst=true to avoid redundancy, meaning if you know all the other categories, you can figure out the last one automatically

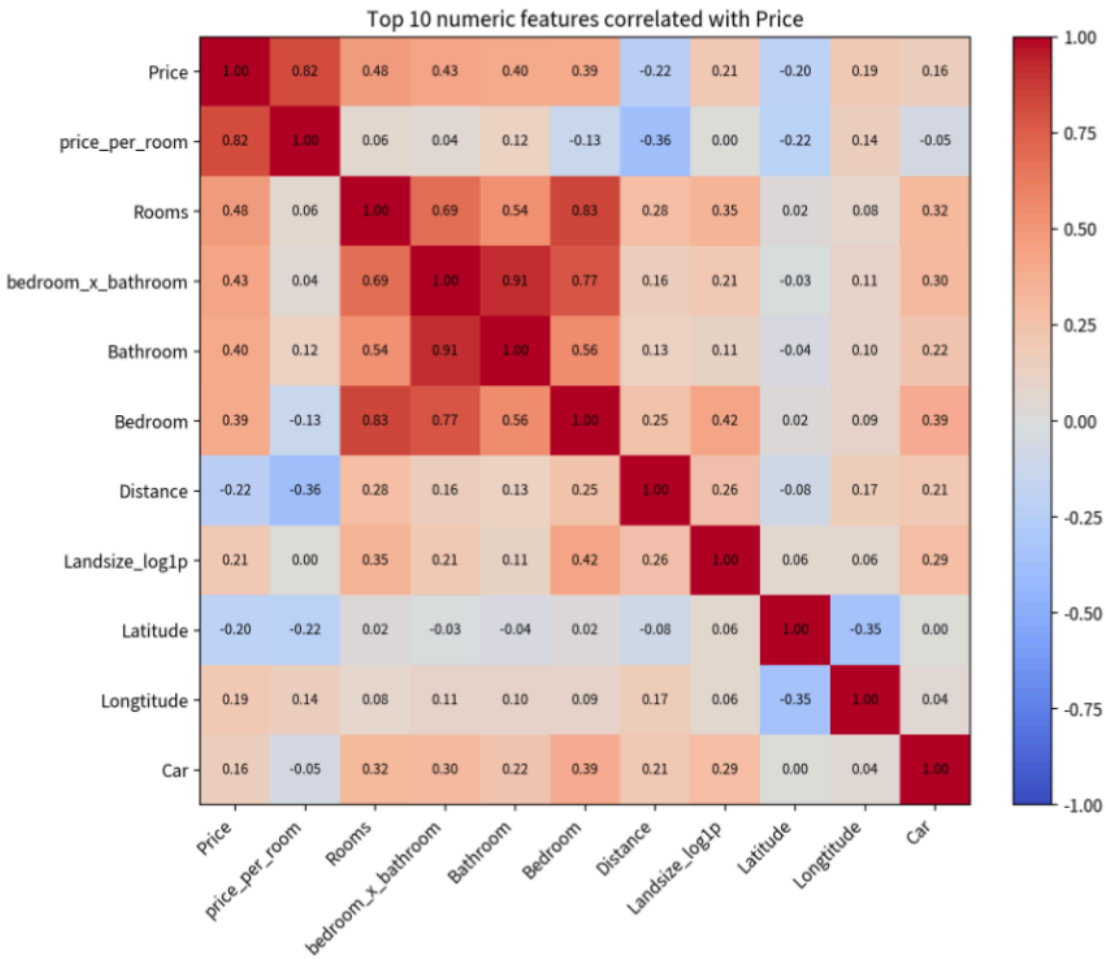
- Ordinal encoding for Regionname

  The region a property is in can tell us a lot about its price. regions closer to the city centre tend to be more expensive than regions further out, so there is a natural order to them. instead of treating all regions as completely separate and equal

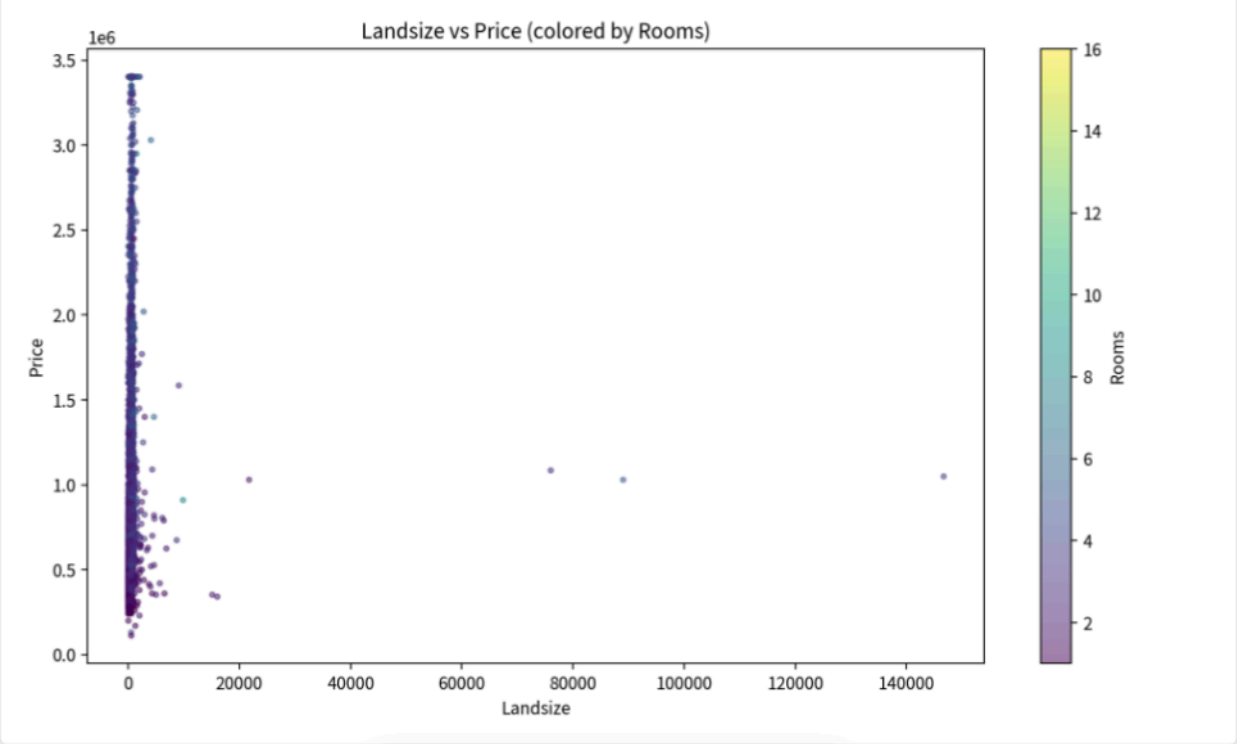
categories with no relationship to each other, i encoded them as an ordered category using CategoricalDtype from the slides. this means each region gets assigned a number based on where it sits in that order, going from the least central regions all the way up to the most central ones. this way the model understands that some regions are "higher" than others in terms of centrality, rather than seeing them all as random unrelated labels.

After all cleaning and feature engineering steps were applied, the final dataset had 18 columns and was fully numeric and ready for analysis.

3.Key findings



- Finding 1. Rooms, bedrooms, and bathrooms are the strongest predictors of price. The correlation heatmap showed that Rooms has the highest correlation with Price (r = 0.51), followed closely by Bedroom2 (r = 0.49) and Bathroom (r = 0.47). This makes sense because buyers are willing to pay significantly more for properties that offer more living space and more bathrooms, and our bedroom_x_bathroom interaction feature was designed to capture exactly this combined effect.
- Finding 2. Property type creates a big gap in average price. The grouped analysis shows that houses sell for an average of $1,247,339, townhouses for $925,163, and units for just $603,907. This means the average house sells for more than double the average unit. This confirms that property type is one of the most important factors in determining price, and encoding it in the right order was a meaningful decision.
- Finding 3. Region creates dramatic price differences across Melbourne. The groupby analysis showed that the average sale price in Southern Metropolitan ($1,338,718) is nearly three and a half times that of Western Victoria ($382,910), even though both are within the same city and state. This shows that location is a critical factor, and a model that ignores region would miss a huge source of price variation across Melbourne.

- Finding 4 — Landsize alone does not strongly predict price. The scatter plot of Landsize vs Price shows that the vast majority of properties are clustered on the left side with small land sizes, and price does not clearly go up as landsize increases. However the color shows that properties with more rooms tend to sit higher on the price axis, which confirms that the number of rooms matters more than the size of the land the property sits on.
- 

4. What I would do next (if this became a full ML project)

- Train some basic models like Linear Regression and test how well they predict price by measuring how far off the predictions are on average.
- Use cross validation to make sure the model performs consistently and not just because we got lucky with how we split the data.
- Handle the Suburb column better since it has too many unique values for regular one hot encoding, so we would need a smarter way to encode it.
- Try more powerful models like Random Forest and Gradient Boosting which tend to work really well on housing data like this.
- Check whether the model performs equally well across all regions in Melbourne or whether it struggles more in some areas than others.

Some parts of this project were developed with the assistance of Artificial intelligence, used to help with debugging code, check column names, and refactoring repeated code into functions. All slide references were researched and identified by me.
All analytical decisions, justifications, and findings are my own